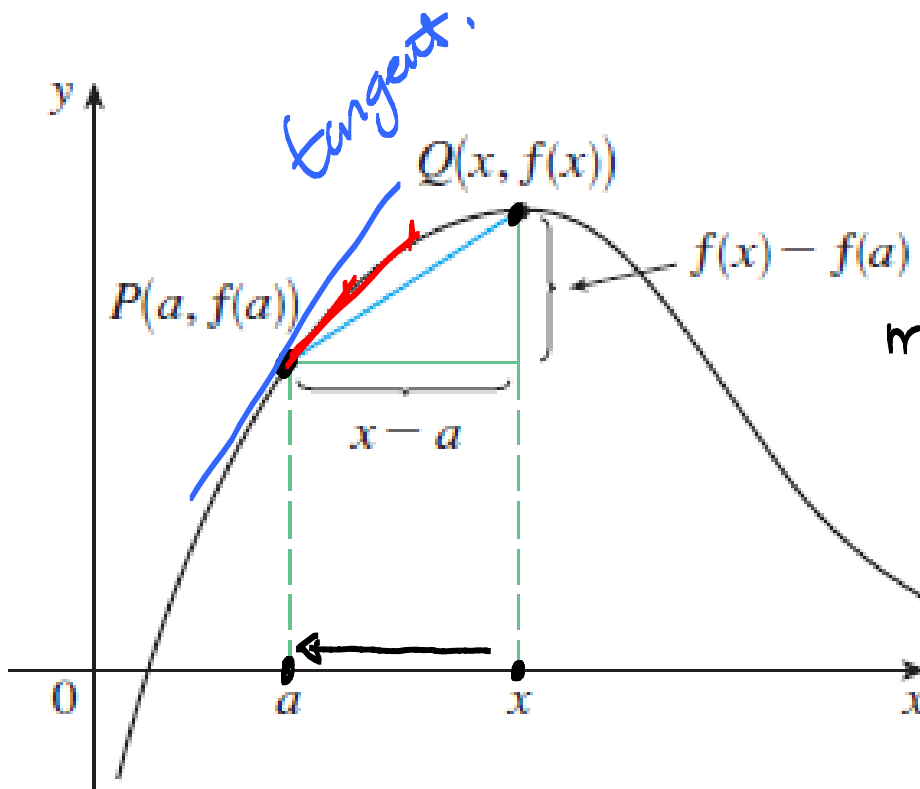




Slope of the
secant PQ

$$m_{PQ} = \frac{f(x) - f(a)}{x - a}$$

Avg rate of
change.

Definition 1: Tangent line to curve $y = f(x)$ at $P(a, f(a))$ is the line through P with slope

$$m = f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided the limit exists!

Example: Find $f'(1)$ and the equation of the tangent line if $f(x) = \frac{1}{x+2}$

$$f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

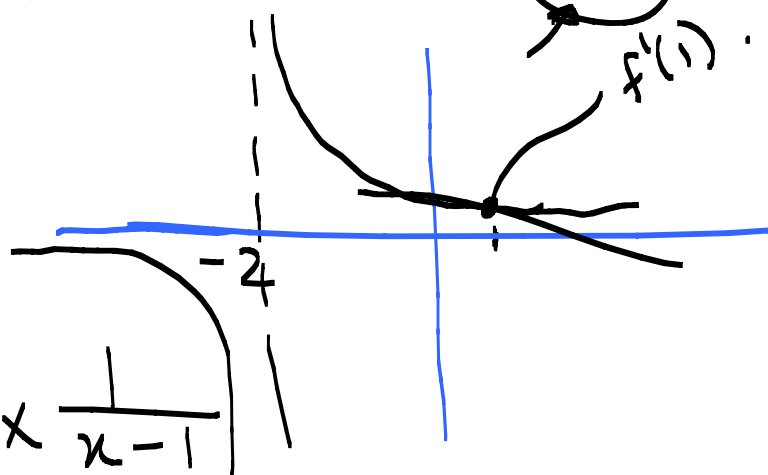
$$= \lim_{x \rightarrow 1} \frac{\frac{1}{x+2} - \frac{1}{3}}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{3 - (x+2)}{3(x+2)} \times \frac{1}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{-x + 1}{3(x+2)(x-1)}$$

$$= \lim_{x \rightarrow 1} \frac{-(x-1)}{3(x+2)(x-1)}$$

$$= -\frac{1}{9}$$



$$(y - y_1) = m(x - x_1)$$

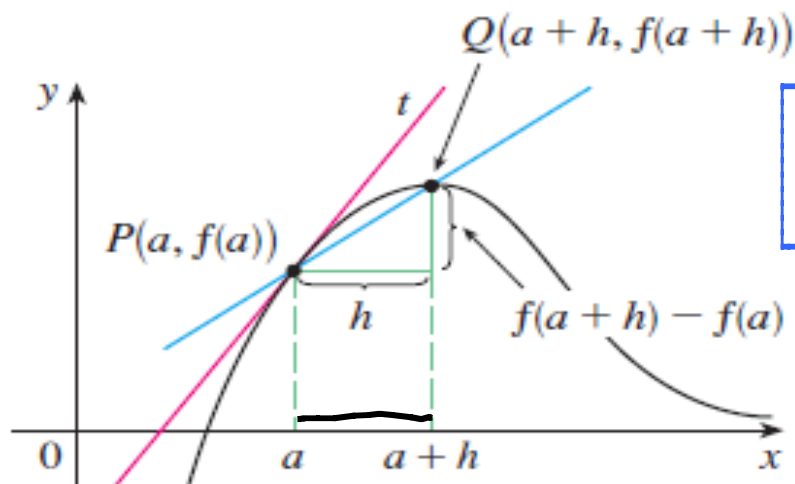
$$m = -\frac{1}{9} \quad (1, \frac{1}{3})$$

$$\Rightarrow \text{eq. of the tang. } y = -\frac{1}{9}x + \frac{4}{9}$$

Check!!

Definition 2:

Alternative expression for the slope of the tangent line.

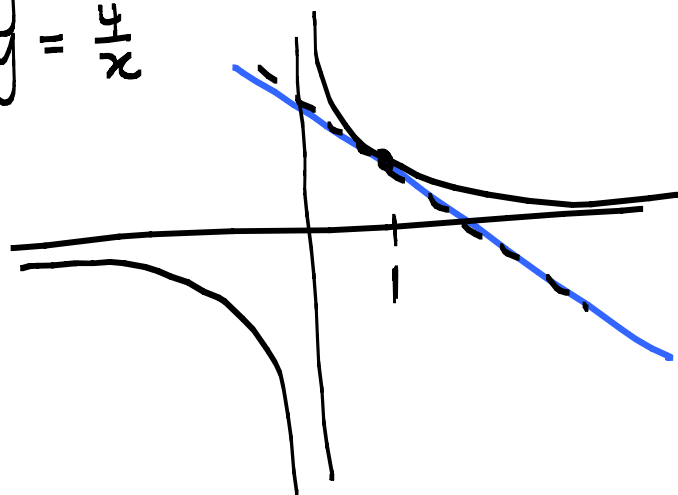


$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

p 142

Example Find the equation of the tangent line to $xy = 4$ where $x=1$.
 $y = \frac{4}{x}$

$$m = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$
$$= -4 \quad (\text{Check!!})$$



First principles / limit definition.

$m = -4$ passing through $(1, 4)$

$\Rightarrow$ Eq of tang is ~~$xy = -4x + 9$~~
 $\checkmark Y = -4x + 8$

Velocities p142 — SELF

Average velocity

Instantaneous velocity

Rates of change p145 — SELF

Average rate of change

Instantaneous rate of change.

Derivatives p 146

4 Definition The derivative of a function f at a number a , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.
~~~~~

OR

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$